192425192

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EXP12:

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#include <stdio.h>
#include <string.h>
#include <ctype.h>

int mod26(int x) { return ((x % 26) + 26) % 26; }

int modinv(int a) {
    int x;
    for (x = 1; x < 26; x++) if ((a * x) % 26 == 1) return x;
    return -1;
}

int main() {
    int K[2][2] = {{9, 4}, {5, 7}};
    int Ki[2][2];
    char raw[] = "meet me at the usual place at ten rather than eight oclock";
    char msg[100], ct[100], pt[100];
    int n = 0, i, det, dinv;

    for (i = 0; raw[i]; i++)
        if (isalpha(raw[i])) msg[n++] = toupper(raw[i]);
    if (n % 2) msg[n++] = 'X';
    msg[n] = '\0';
    printf("Plaintext : %s\n", msg);
    printf("Key matrix: [%d %d; %d %d]\n\n", K[0][0], K[0][1], K[1][0], K[1][1]);

    printf("Encryption (C = K * P mod 26):\n");
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    for (i = 0; i < n; i += 2) {
        int p1 = msg[i] - 'A', p2 = msg[i + 1] - 'A';
        int c1 = mod26(K[0][0] * p1 + K[0][1] * p2);
        int c2 = mod26(K[1][0] * p1 + K[1][1] * p2);
        ct[i] = c1 + 'A'; ct[i + 1] = c2 + 'A';
        printf("  %c%c = (%2d,%2d) -> (%d*%d+%d*%d, %d*%d+%d*%d) mod 26 = (%2d, %2d) = %c%c\n",
                msg[i], msg[i + 1], p1, p2, K[0][0], p1, K[0][1], p2,
                K[1][0], p1, K[1][1], p2, c1, c2, ct[i], ct[i + 1]);
    }
    ct[n] = '\0';
    printf("Ciphertext: %s\n\n", ct);

    det = mod26(K[0][0] * K[1][1] - K[0][1] * K[1][0]);
    dinv = modinv(det);
    Ki[0][0] = mod26(dinv * K[1][1]);   Ki[0][1] = mod26(-dinv * K[0][1]);
    Ki[1][0] = mod26(-dinv * K[1][0]);  Ki[1][1] = mod26(dinv * K[0][0]);
    printf("det(K) mod 26 = %d, inverse of det = %d\n", det, dinv);
    printf("Inverse key: [%d %d; %d %d]\n\n", Ki[0][0], Ki[0][1], Ki[1][0], Ki[1][1]);

    printf("Decryption (P = K^-1 * C mod 26):\n");
    for (i = 0; i < n; i += 2) {
        int c1 = ct[i] - 'A', c2 = ct[i + 1] - 'A';
        int p1 = mod26(Ki[0][0] * c1 + Ki[0][1] * c2);
        int p2 = mod26(Ki[1][0] * c1 + Ki[1][1] * c2);
        pt[i] = p1 + 'a'; pt[i + 1] = p2 + 'a';
        printf("  %c%c = (%2d,%2d) -> (%2d,%2d) = %c%c\n", ct[i], ct[i + 1], c1, c2, p1, p2, pt[i], pt[i + 1]);
    }
    pt[n] = '\0';
    printf("Recovered plaintext: %s\n", pt);
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    return 0;
}
```

OUTPUT:

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Plaintext : MEETMEATTHEUSUALPLACEATTENRATHERTHANEIGHTOCLOCKX
Key matrix: [9 4; 5 7]

Encryption (C = K * P mod 26):
  ME = (12, 4) -> (9*12+4*4, 5*12+7*4) mod 26 = (20,10) = UK
  ET = ( 4,19) -> (9*4+4*19, 5*4+7*19) mod 26 = ( 8,23) = IX
  ME = (12, 4) -> (9*12+4*4, 5*12+7*4) mod 26 = (20,10) = UK
  AT = ( 0,19) -> (9*0+4*19, 5*0+7*19) mod 26 = (24, 3) = YD
  TH = (19, 7) -> (9*19+4*7, 5*19+7*7) mod 26 = (17,14) = RO
  EU = ( 4,20) -> (9*4+4*20, 5*4+7*20) mod 26 = (12, 4) = ME
  SU = (18,20) -> (9*18+4*20, 5*18+7*20) mod 26 = ( 8,22) = IW
  AL = ( 0,11) -> (9*0+4*11, 5*0+7*11) mod 26 = (18,25) = SZ
  PL = (15,11) -> (9*15+4*11, 5*15+7*11) mod 26 = (23,22) = XW
  AC = ( 0, 2) -> (9*0+4*2, 5*0+7*2) mod 26 = ( 8,14) = IO
  EA = ( 4, 0) -> (9*4+4*0, 5*4+7*0) mod 26 = (10,20) = KU
  TT = (19,19) -> (9*19+4*19, 5*19+7*19) mod 26 = (13,20) = NU
  EN = ( 4,13) -> (9*4+4*13, 5*4+7*13) mod 26 = (10, 7) = KH
  RA = (17, 0) -> (9*17+4*0, 5*17+7*0) mod 26 = (23, 7) = XH
  TH = (19, 7) -> (9*19+4*7, 5*19+7*7) mod 26 = (17,14) = RO
  ER = ( 4,17) -> (9*4+4*17, 5*4+7*17) mod 26 = ( 0, 9) = AJ
  TH = (19, 7) -> (9*19+4*7, 5*19+7*7) mod 26 = (17,14) = RO
  AN = ( 0,13) -> (9*0+4*13, 5*0+7*13) mod 26 = ( 0,13) = AN
  EI = ( 4, 8) -> (9*4+4*8, 5*4+7*8) mod 26 = (16,24) = QY
  GH = ( 6, 7) -> (9*6+4*7, 5*6+7*7) mod 26 = ( 4, 1) = EB
  TO = (19,14) -> (9*19+4*14, 5*19+7*14) mod 26 = (19,11) = TL
  CL = ( 2,11) -> (9*2+4*11, 5*2+7*11) mod 26 = (10, 9) = KJ
  OC = (14, 2) -> (9*14+4*2, 5*14+7*2) mod 26 = ( 4, 6) = EG
  KX = (10,23) -> (9*10+4*23, 5*10+7*23) mod 26 = ( 0, 3) = AD
Ciphertext: UKIXUKYDROMEIWSZXWIOKUNUKHXHROAJROANQYEBTLKJEGAD

det(K) mod 26 = 17, inverse of det = 23
Inverse key: [5 12; 15 25]

Decryption (P = K^-1 * C mod 26):
  UK = (20,10) -> (12, 4) = me
  IX = ( 8,23) -> ( 4,19) = et
  UK = (20,10) -> (12, 4) = me
  YD = (24, 3) -> ( 0,19) = at
  RO = (17,14) -> (19, 7) = th
  ME = (12, 4) -> ( 4,20) = eu
  IW = ( 8,22) -> (18,20) = su
  SZ = (18,25) -> ( 0,11) = al
  XW = (23,22) -> (15,11) = pl
  IO = ( 8,14) -> ( 0, 2) = ac
  KU = (10,20) -> ( 4, 0) = ea
  NU = (13,20) -> (19,19) = tt
  KH = (10, 7) -> ( 4,13) = en
  XH = (23, 7) -> (17, 0) = ra
  RO = (17,14) -> (19, 7) = th
  AJ = ( 0, 9) -> ( 4,17) = er
  RO = (17,14) -> (19, 7) = th
  AN = ( 0,13) -> ( 0,13) = an
  QY = (16,24) -> ( 4, 8) = ei
  EB = ( 4, 1) -> ( 6, 7) = gh
  TL = (19,11) -> (19,14) = to
  KJ = (10, 9) -> ( 2,11) = cl
  EG = ( 4, 6) -> (14, 2) = oc
  AD = ( 0, 3) -> (10,23) = kx
Recovered plaintext: meetmeattheusualplaceattenratherthaneightoclockx
```